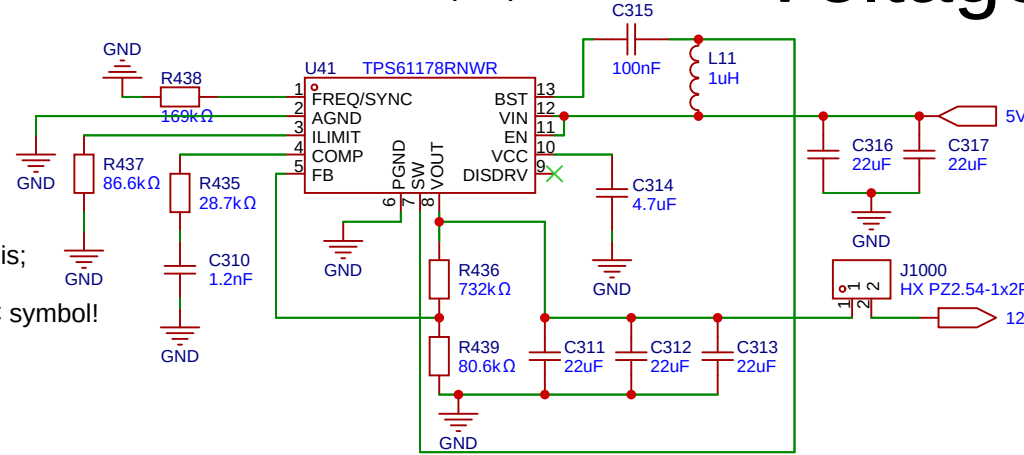
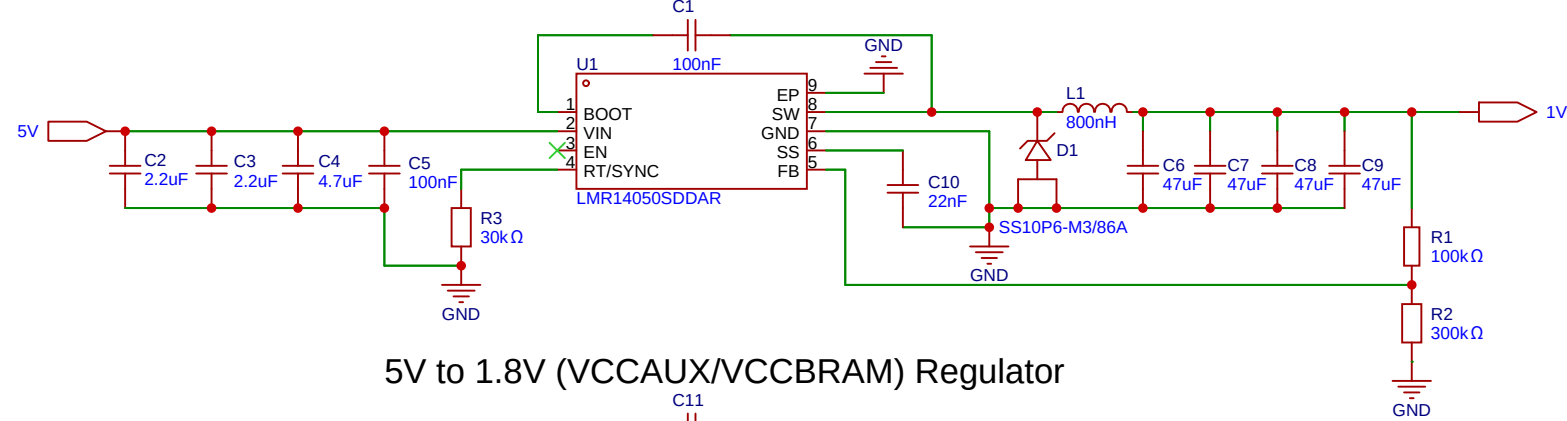


Sorry about how terrible this layout is;
it was the best I could do given the
way they laid out the pins on this IC symbol!

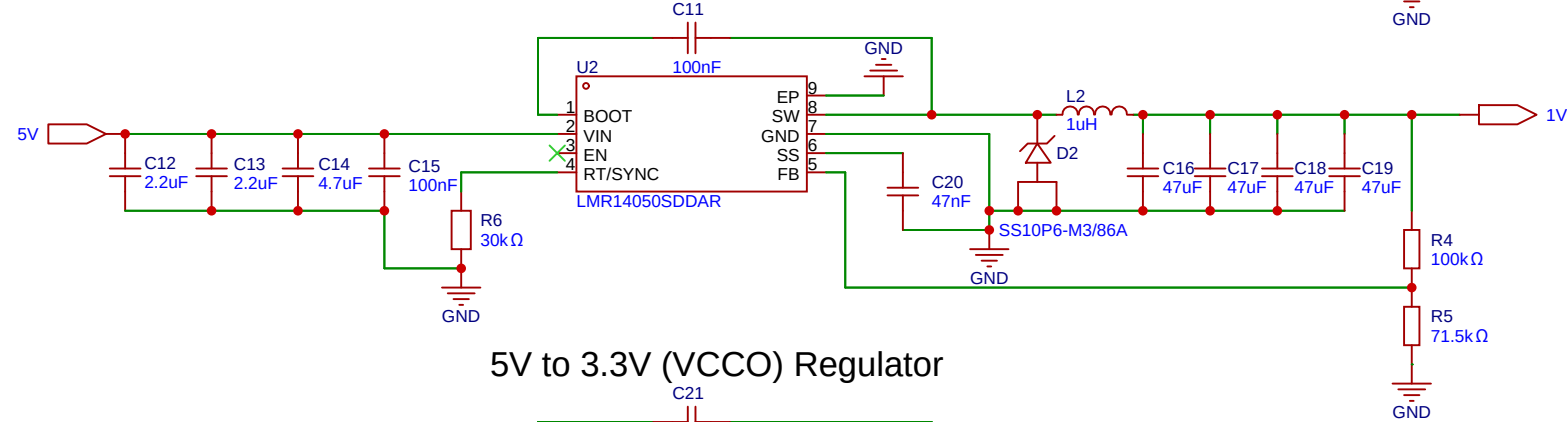
5V to 12V Step-Up Converter



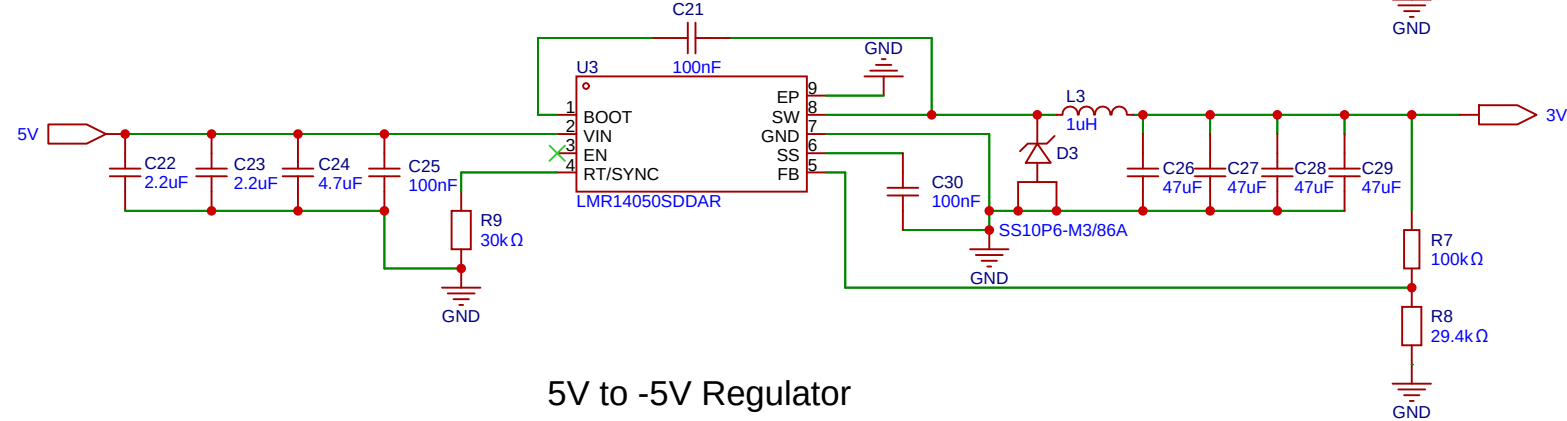
5V to 1V (VCCINT) Regulator



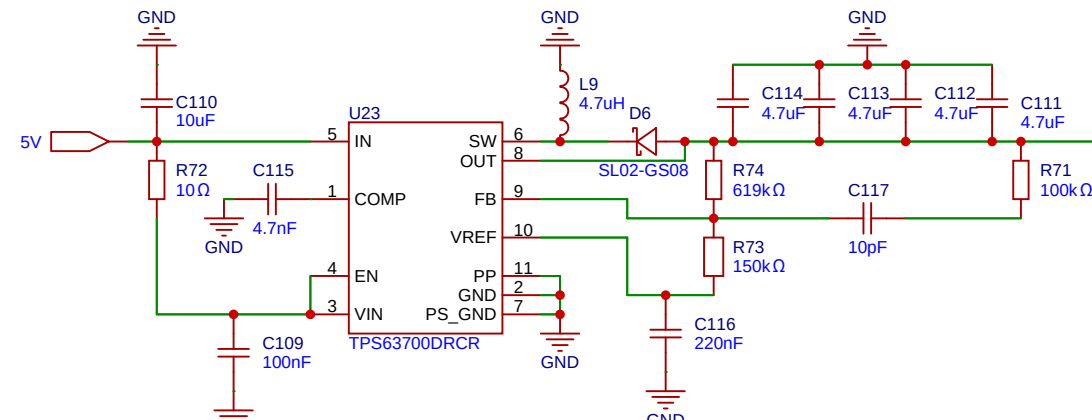
5V to 1.8V (VCCAUX/VCCBRAM) Regulator



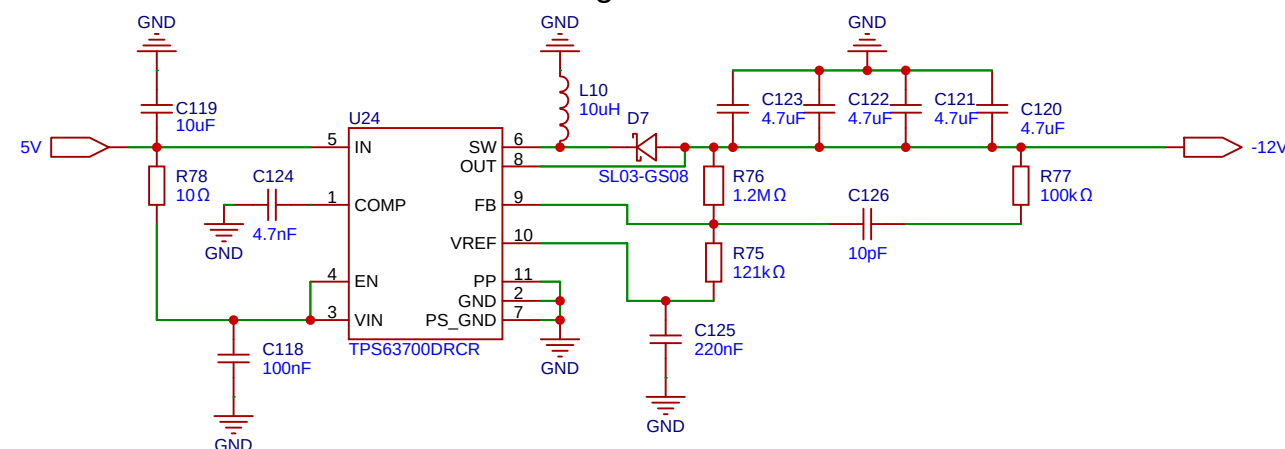
5V to 3.3V (VCCO) Regulator



5V to -5V Regulator

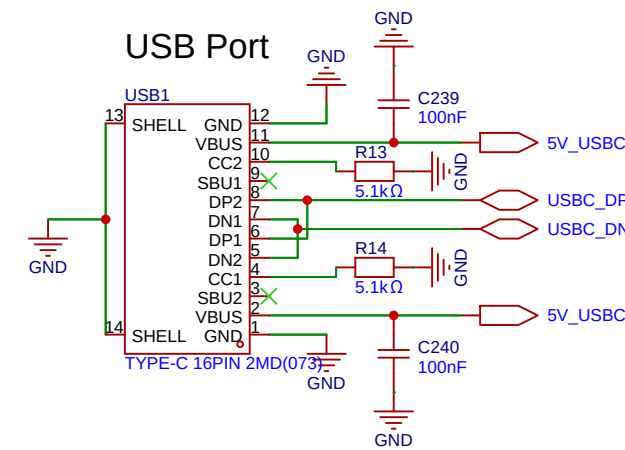


5V to -12V Regulator

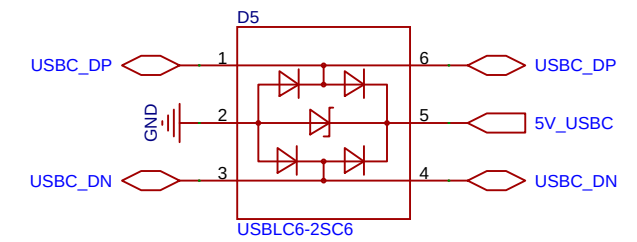


Voltage Regulation

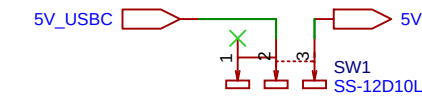
USB Port and Power Switching



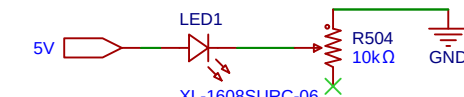
ESD Suppressor



Power Switch

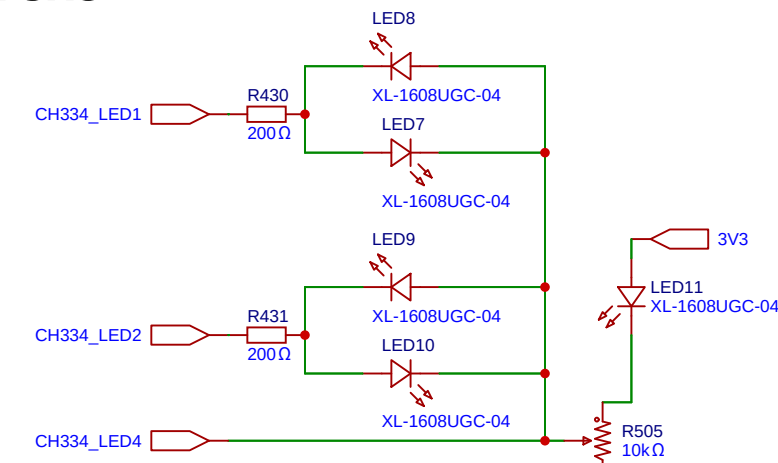


Power LED

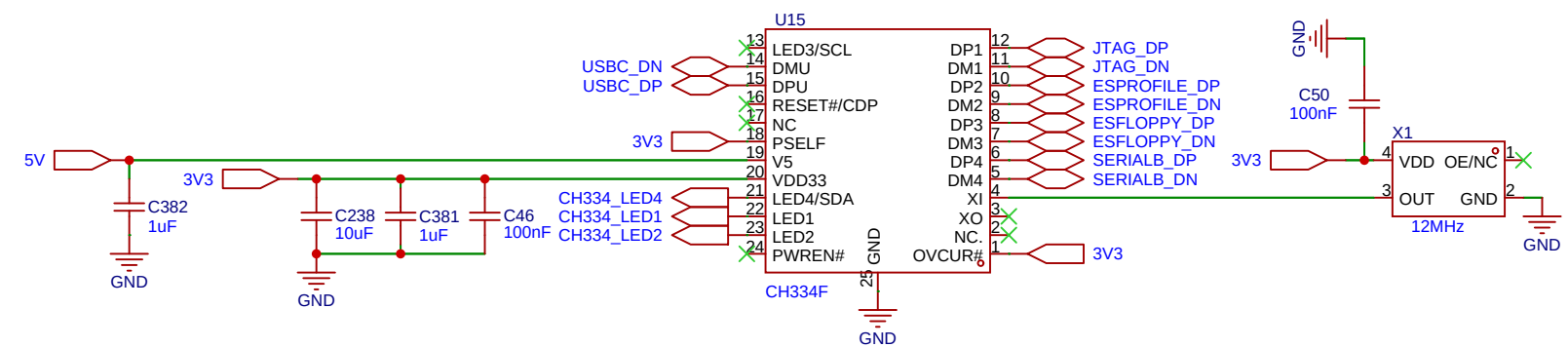


4-Port USB Hub

Port Status LEDs

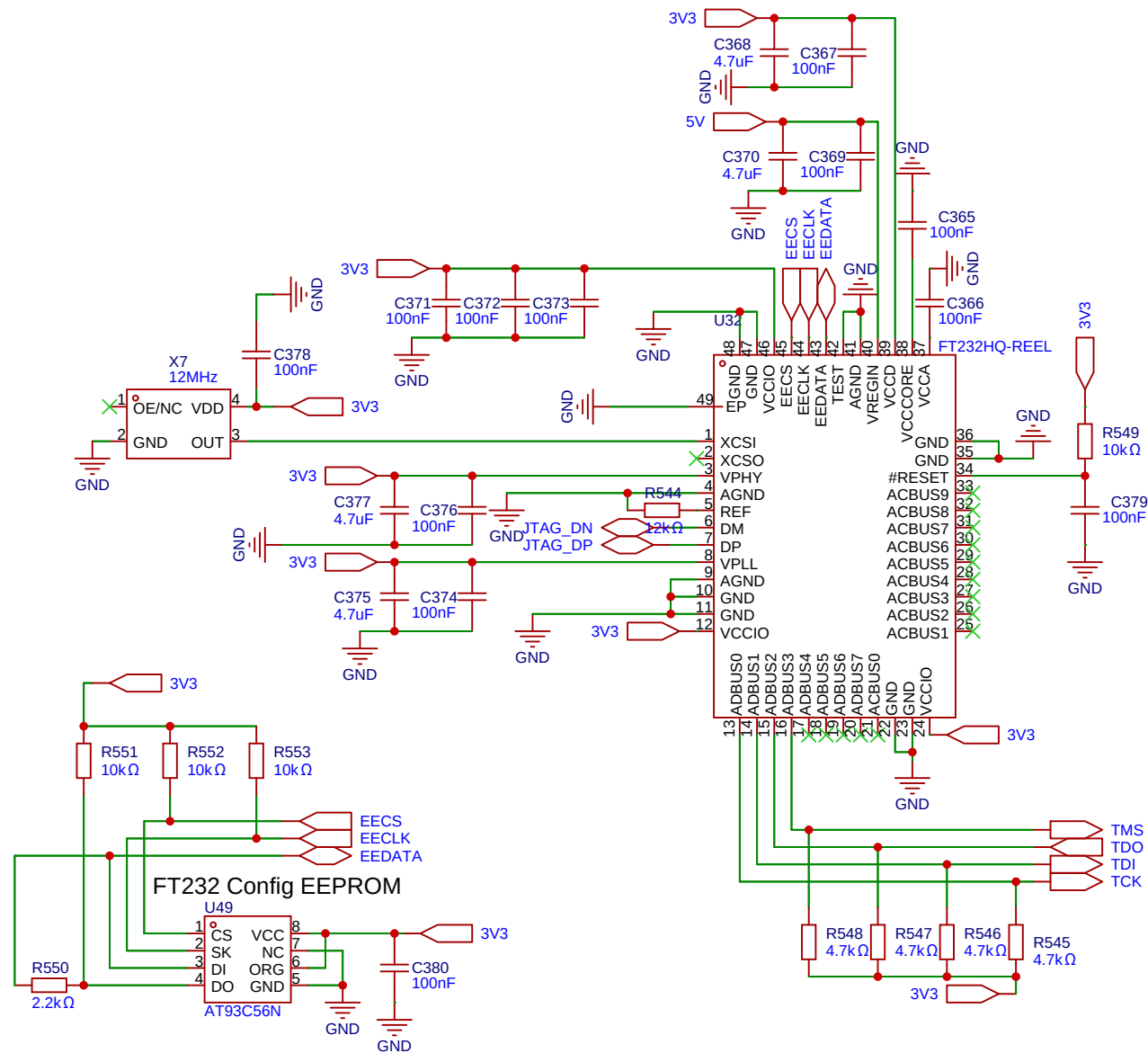


CH334F USB Hub

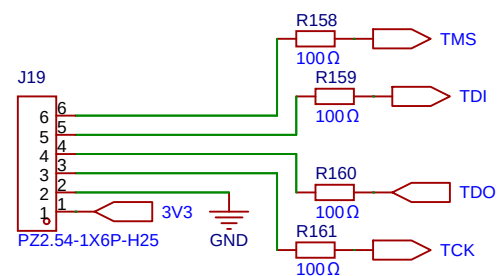


Schematic	LisaFPGA Schematic			Create at	2025-10-14
				Update at	2026-02-01
Board	LisaFPGA			Page	Power and USB Input
Drawn		LisaFPGA			
Reviewed					
		Version	Size	Page 1 Total 8	
		V1.0	A4	EasyEDA.com	

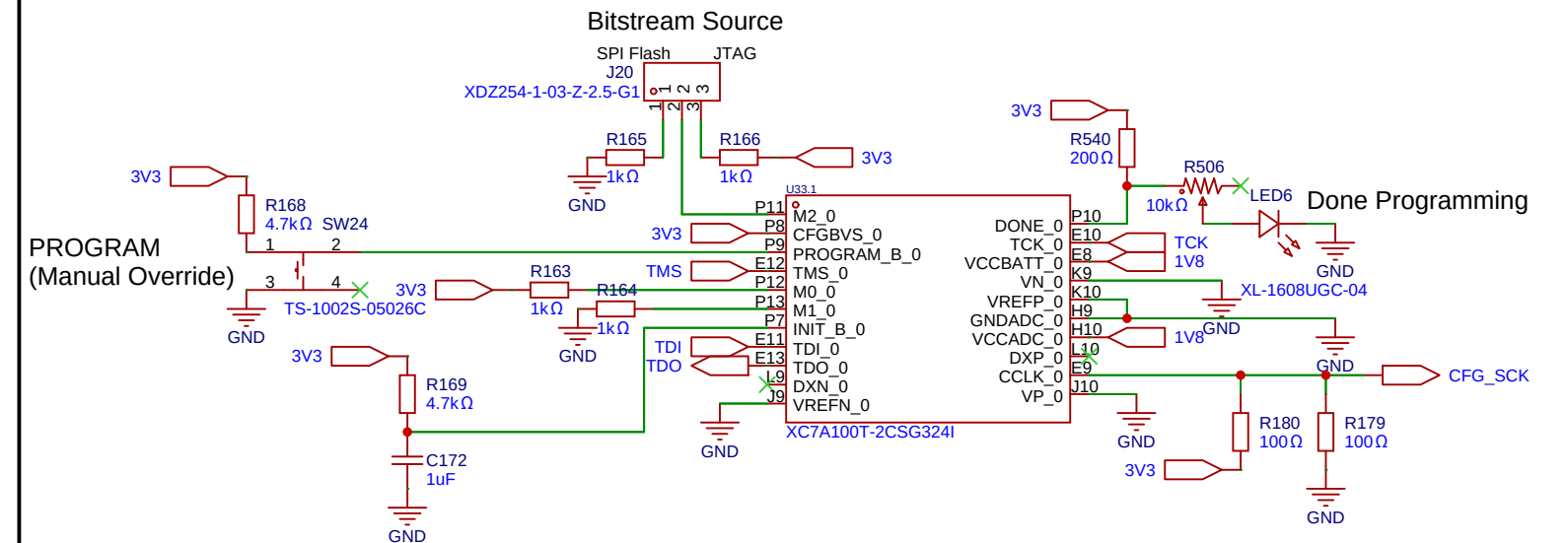
FT232 USB to JTAG



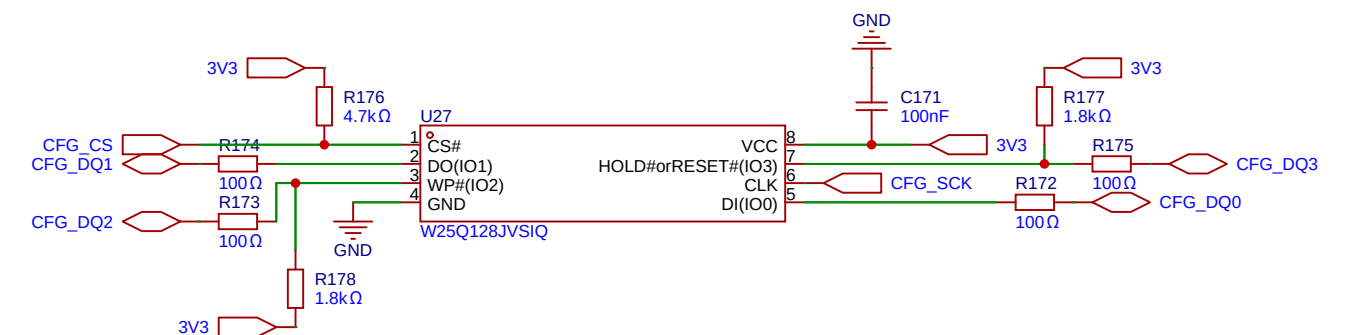
Dedicated JTAG Header



FPGA Programming/Config Interface

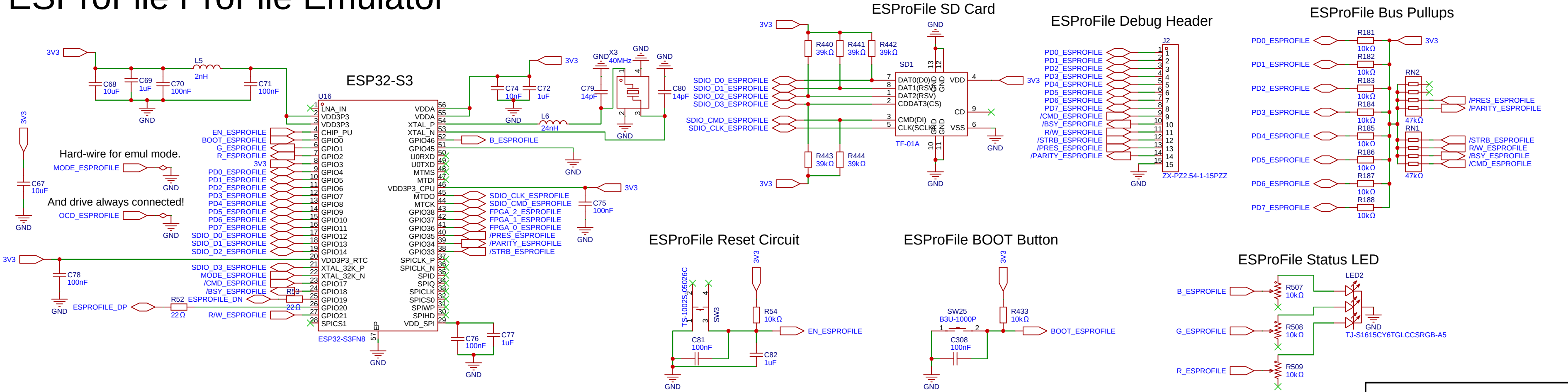


Bitstream SPI Flash

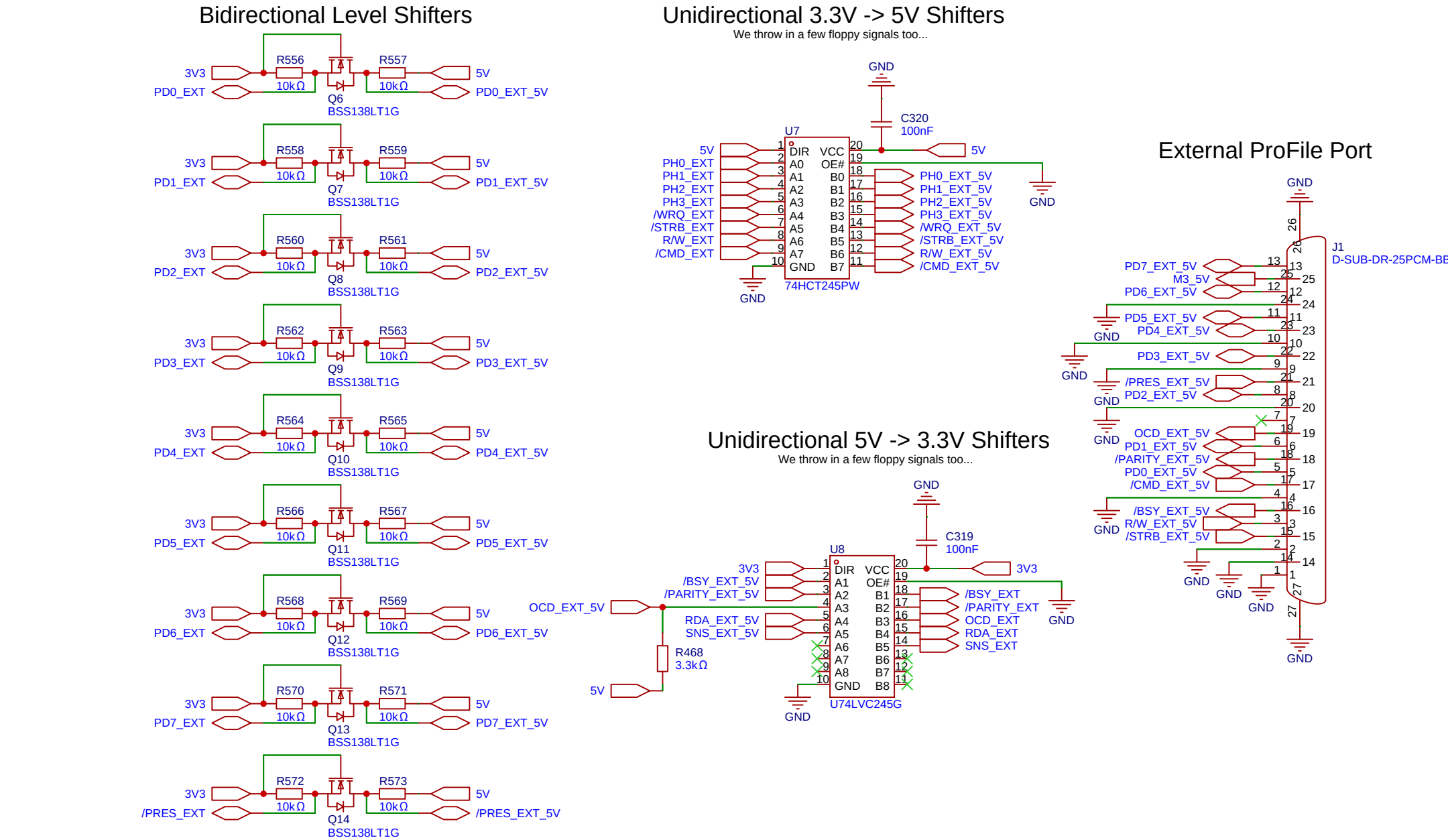


Schematic	LisaFPGA Schematic			Create at	2025-10-15
				Update at	2026-02-01
Board	LisaFPGA			Page	FPGA Programming
Drawn		LisaFPGA			
Reviewed					
		Version	Size	Page 2 Total 8	
		V1.0	A4	EasyEDA.com	

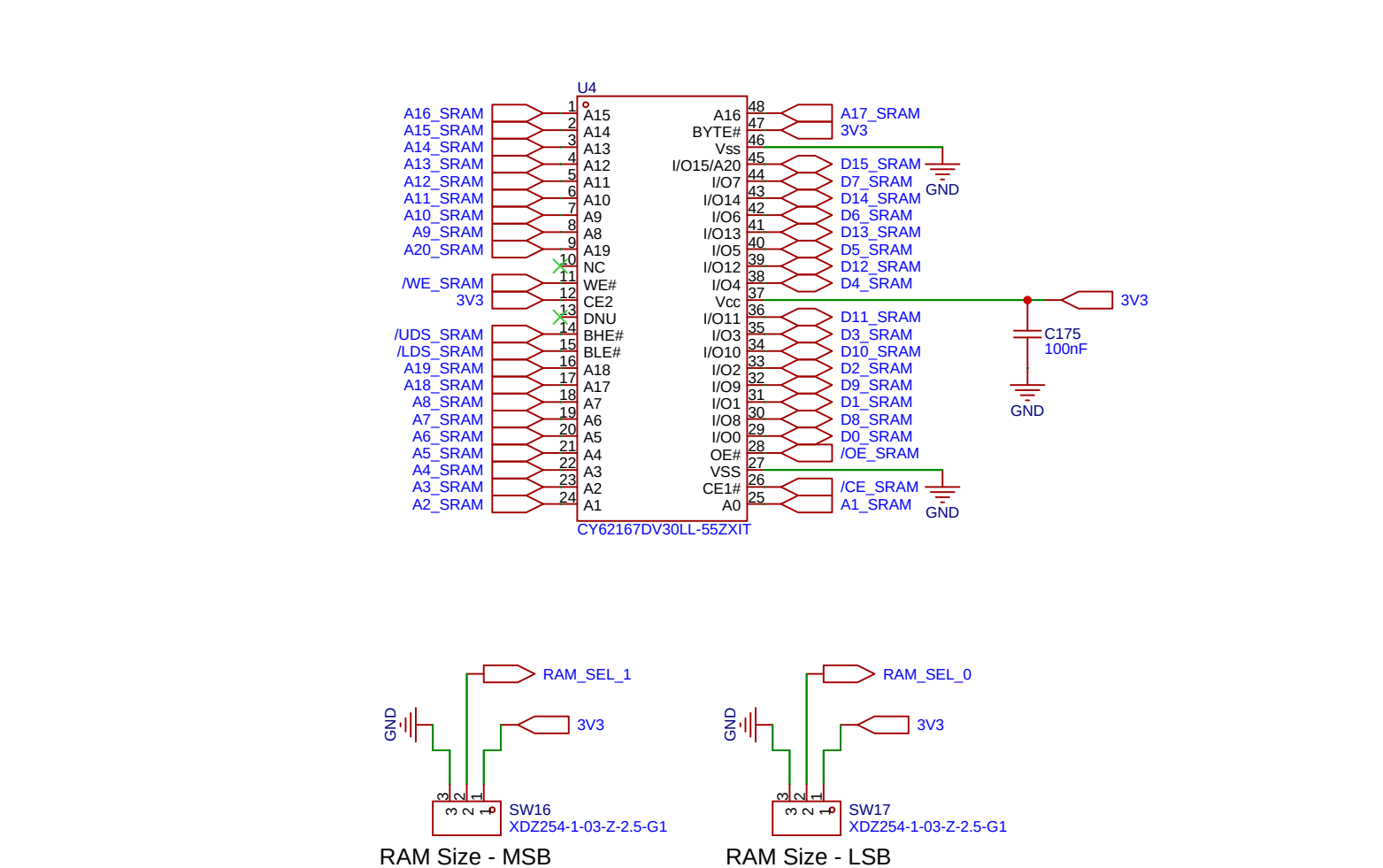
ESProFile ProFile Emulator



External ProFile Connection

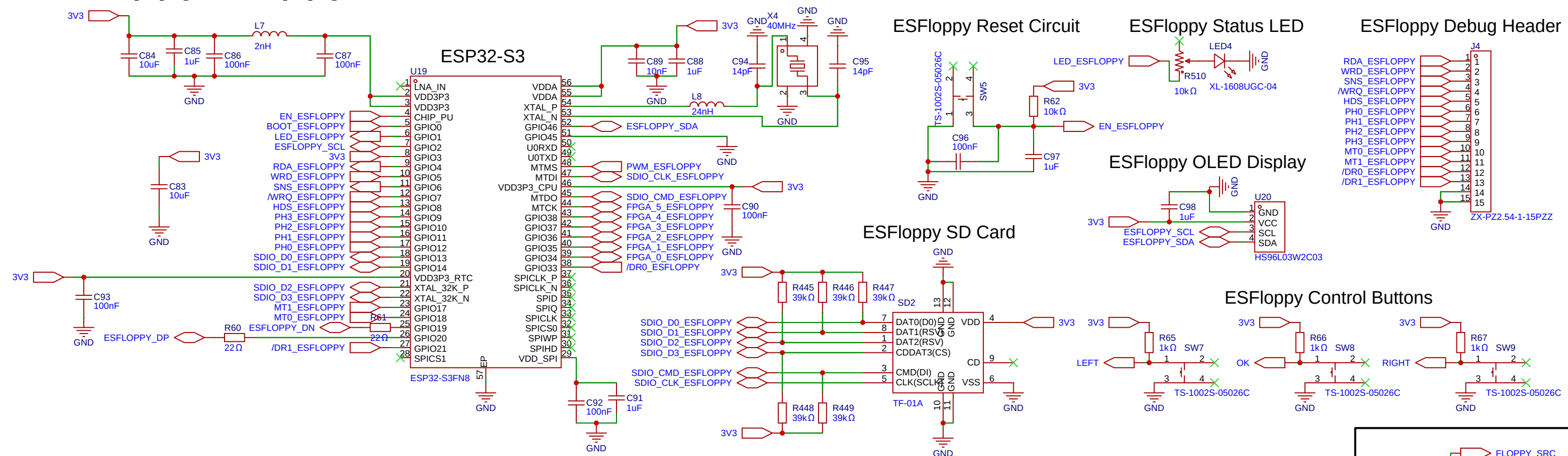


Lisa System Memory

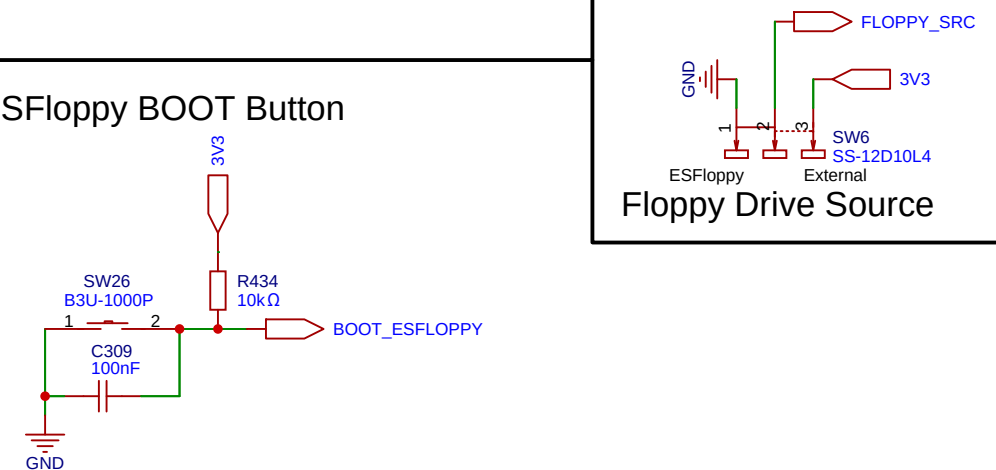
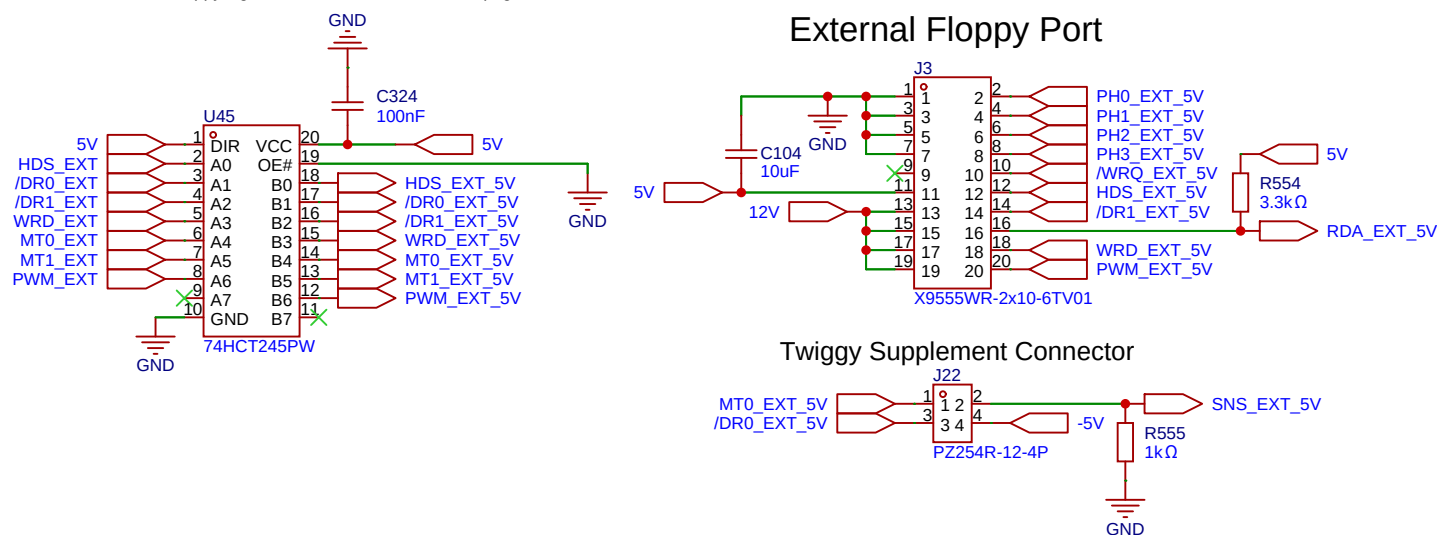


Schematic	LisaFPGA Schematic		Create at	2025-10-24
Board	LisaFPGA		Update at	2026-02-01
Drawn			Page	ProFile Interface and SRAM
Reviewed			LisaFPGA	
		Version	Size	Page 3 Total 8
		V1.0	A4	EasyEDA.com

ESFloppy Floppy Emulator



External Floppy Connection

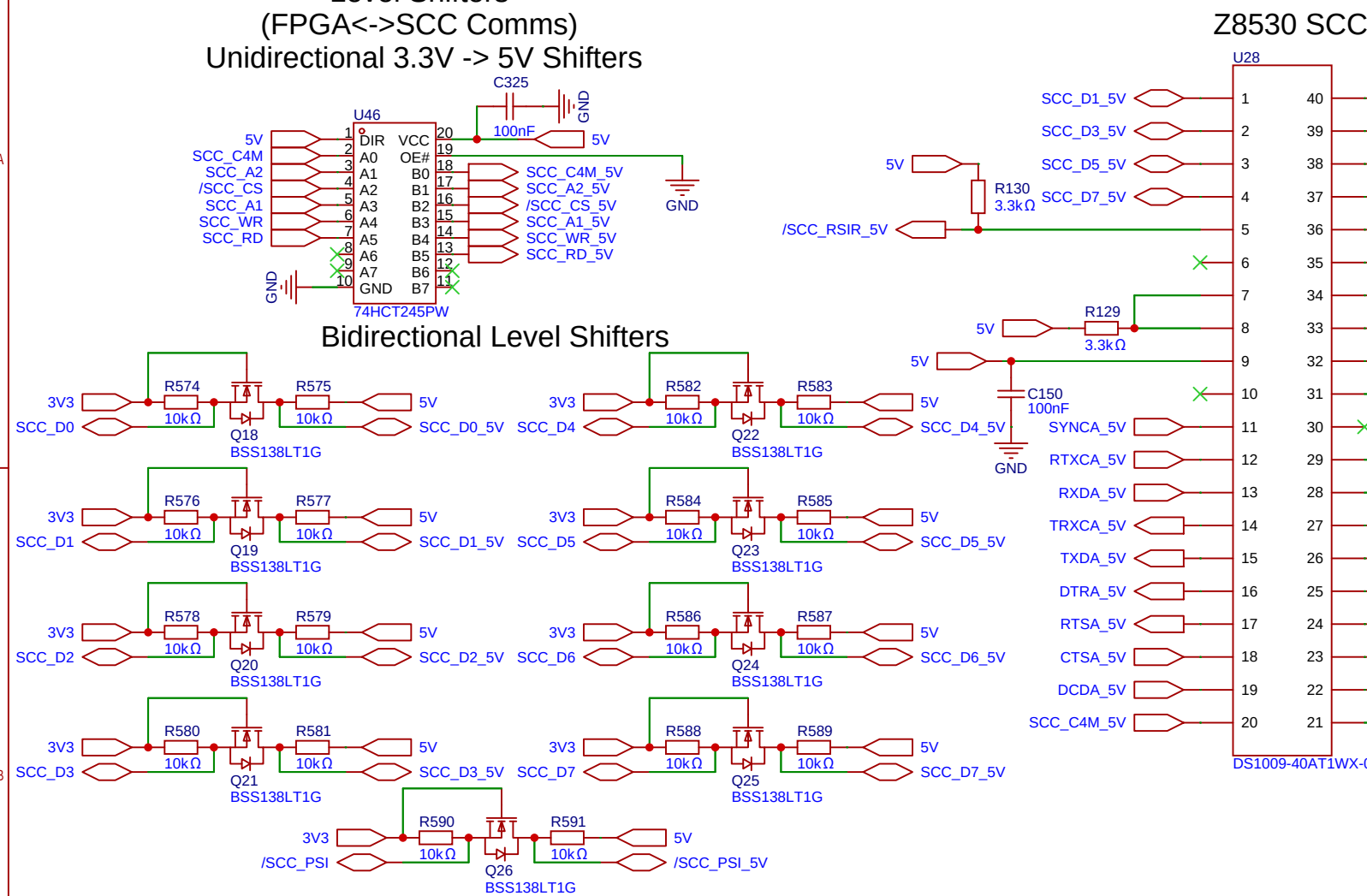


I ran out of space up there,
so the BOOT button has to go here.
All alone in this massive space.
That was some really poor planning on my part.
Whoops.

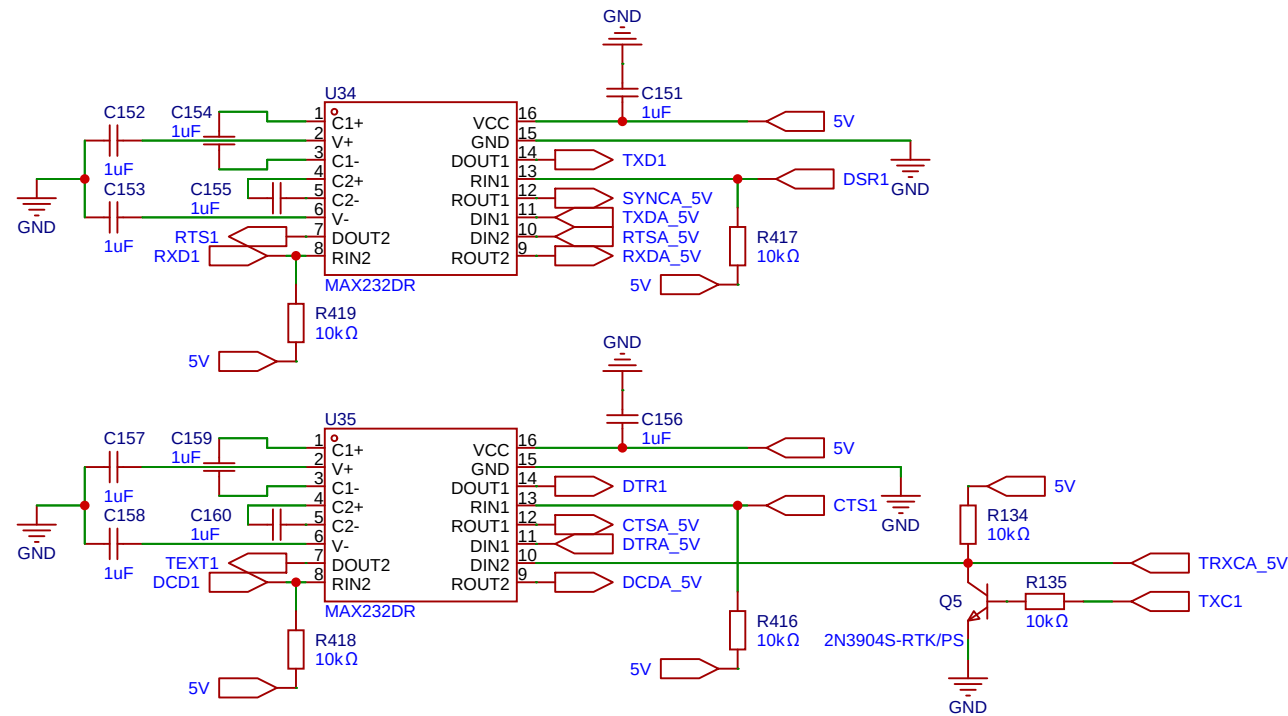
Schematic	LisaFPGA Schematic			Create at	2025-10-27
				Update at	2026-02-01
Board	LisaFPGA			Page	Floppy Interface
Drawn		LisaFPGA			
Reviewed					
		Version	Size	Page 4 Total 8	
		V1.0	A4	EasyEDA.com	

The SCC Itself!

Level Shifters
(FPGA<->SCC Comms)
Unidirectional 3.3V -> 5V Shifters

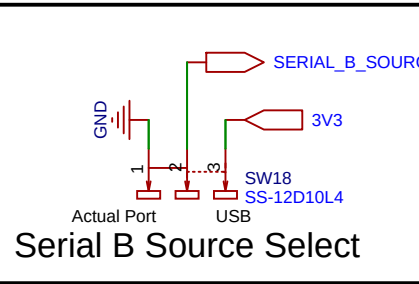
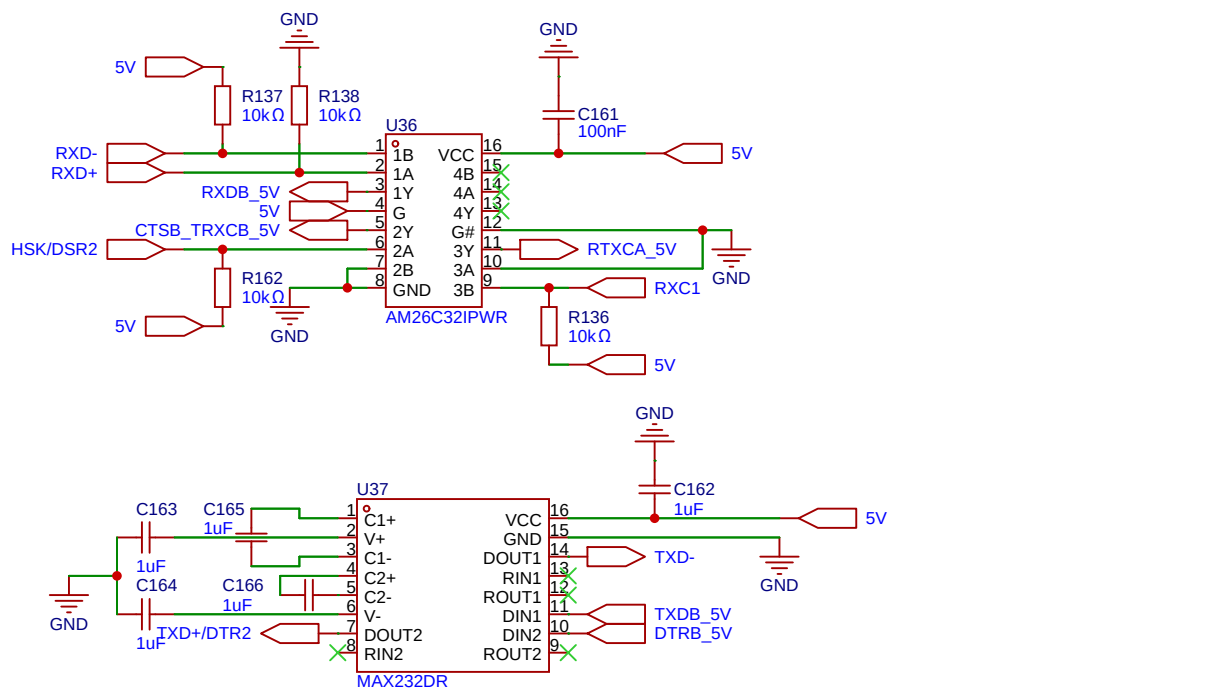


Serial A Transceivers



Serial B Transceivers

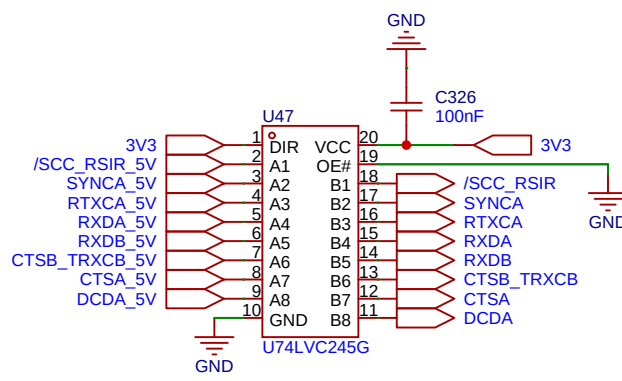
A note about Serial B: THE TXD LINES DON'T SUPPORT DIFFERENTIAL OPERATION!
Serial B can operate in 2 modes: differential (RS-422) and single-ended (RS-232).
But we can't do differential because I couldn't find a modern line driver IC that could be switched between differential and single-ended on the fly like the Lisa requires.
So I instead used a single-ended IC drivers since you'll be using the port in RS-232 mode 99 times out of 100 anyway.
The RXD lines do still support differential mode though, so at least you can receive differential data!



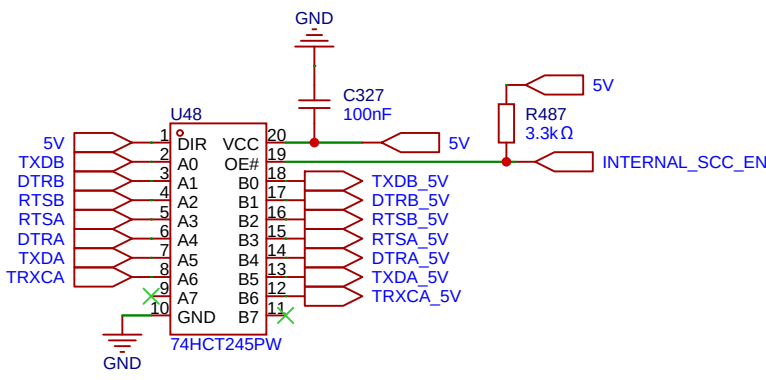
Serial I/O to FPGA

Level Shifters
(FPGA Direct Drive of Serial Lines)
Useless until we have a Verilog SCC!
Plus RSIR, which is actually useful!

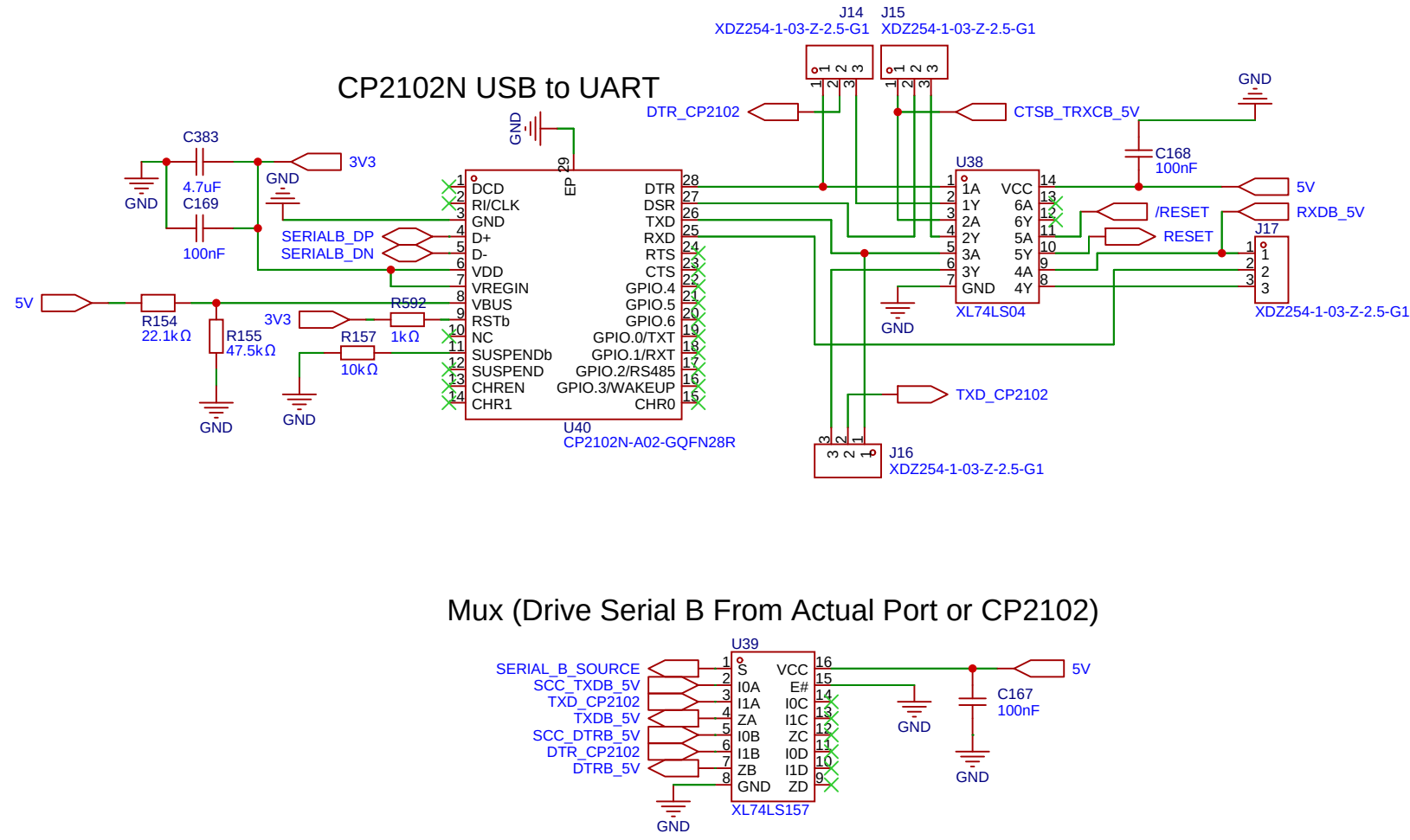
Unidirectional 5V -> 3.3V Shifters



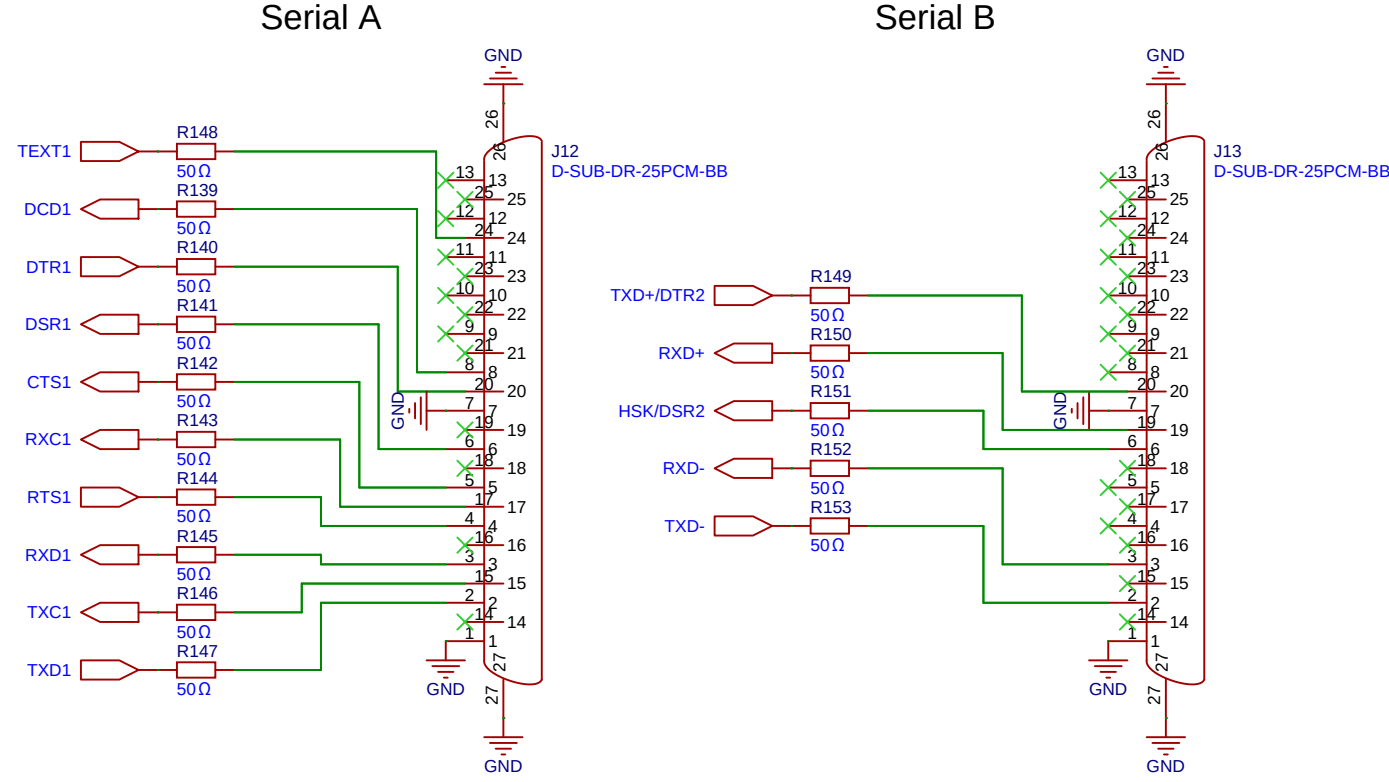
Unidirectional 3.3V -> 5V Shifters



USB to Serial B

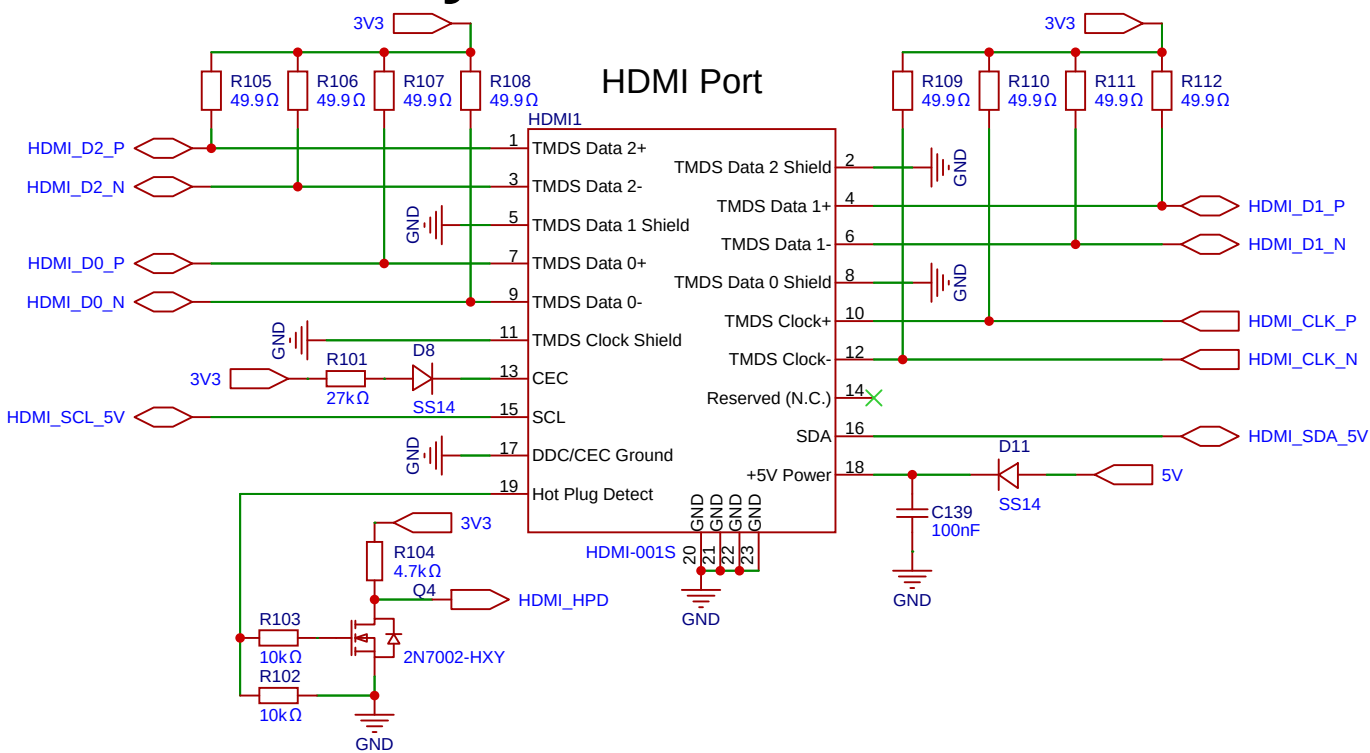


Physical Ports

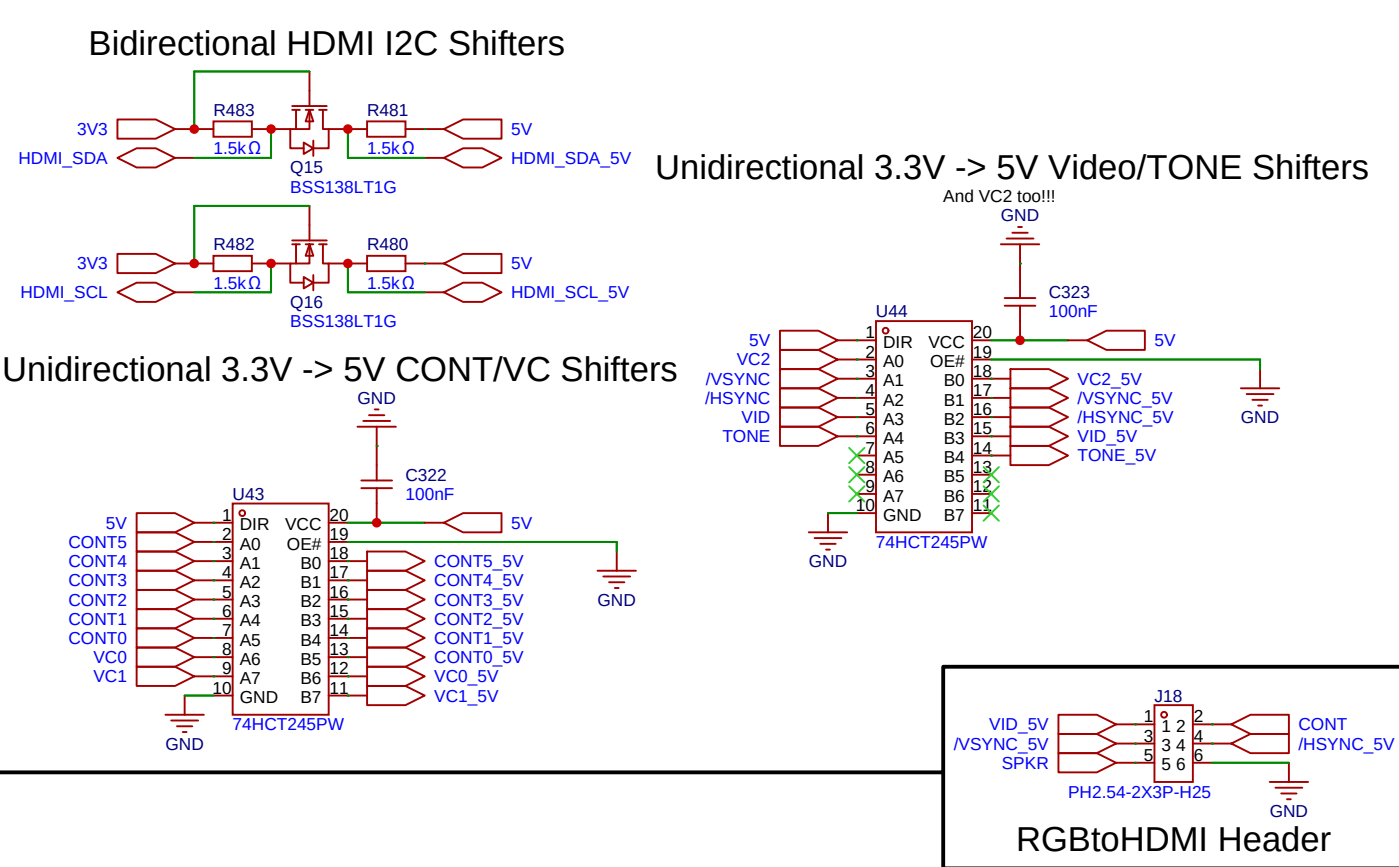


Schematic	LisaFPGA Schematic		Create at	2025-10-28
Board	LisaFPGA		Update at	2026-02-01
Drawn			Page	Serial Ports
Reviewed			LisaFPGA	
		Version	Size	Page 5 Total 8
EasyEDA		V1.0	A4	EasyEDA.com

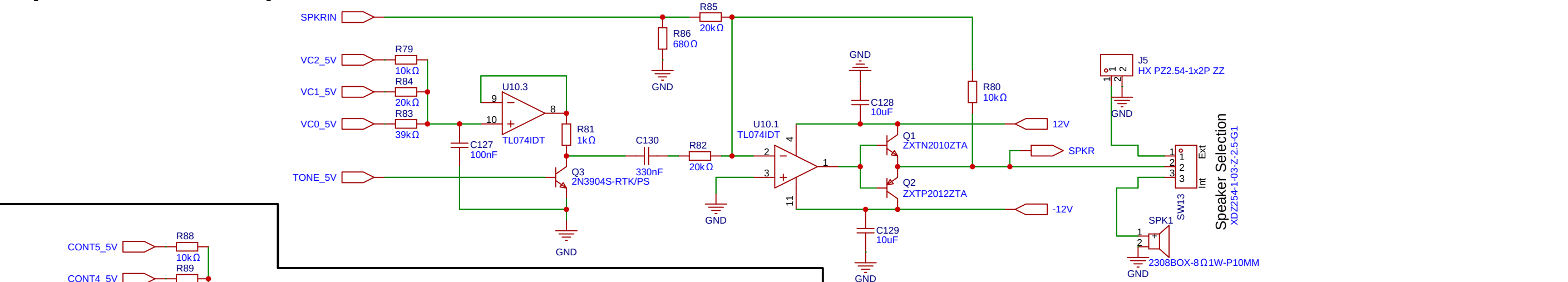
HDMI Subsystem



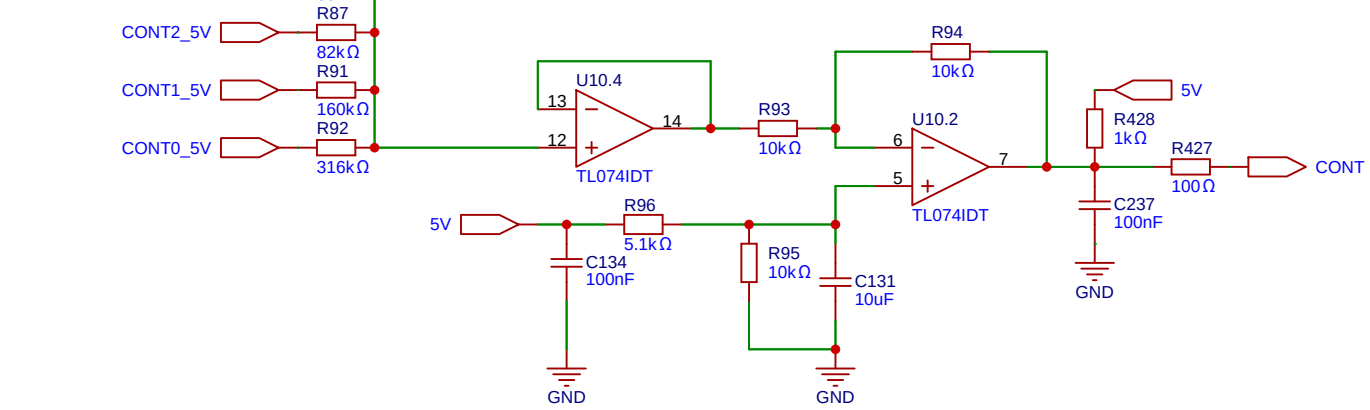
Level Shifters



Speaker Amp



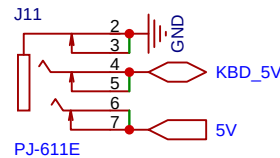
Contrast DAC



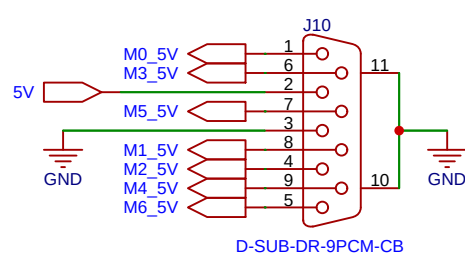
Schematic	LisaFPGA Schematic			Create at	2025-10-28
Board	LisaFPGA			Update at	2026-01-08
Drawn				Page	Audio/Video
Reviewed				LisaFPGA	
		Version	Size	Page 6 Total 8	
		V1.0	A4	EasyEDA.com	

Keyboard/Mouse

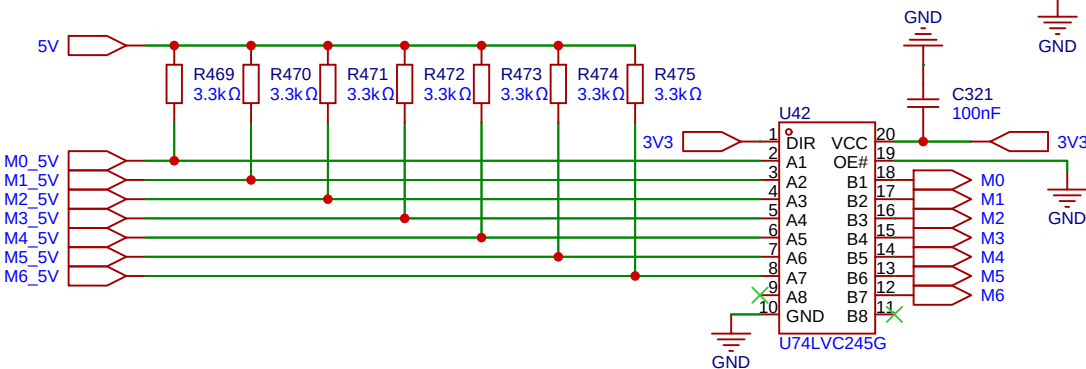
Lisa Keyboard Port



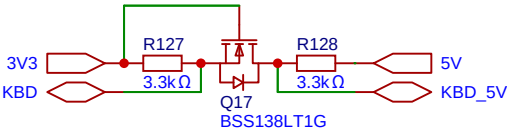
Lisa Mouse Port



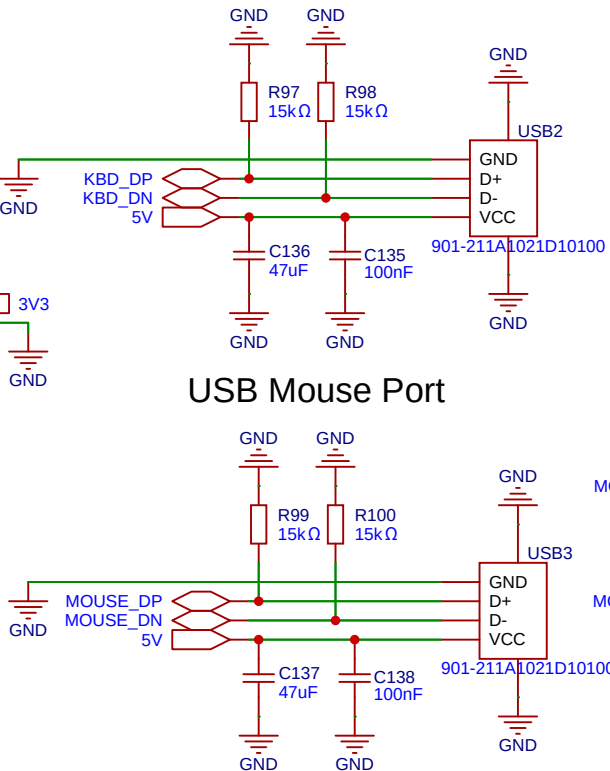
Unidirectional 5V -> 3.3V Lisa Mouse Shifters



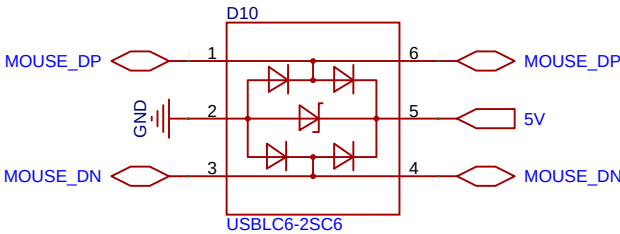
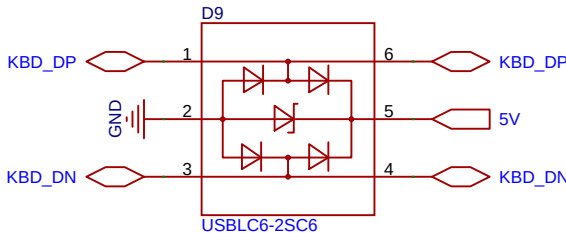
Bidirectional Lisa Keyboard Level Shifter



USB Mouse Port

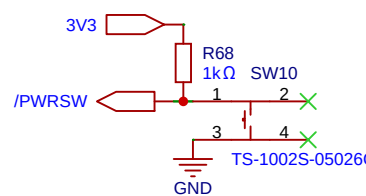


ESD Suppressor

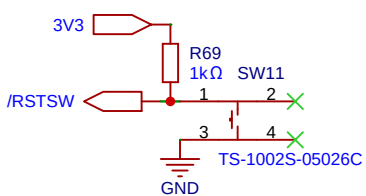


Lisa Buttons

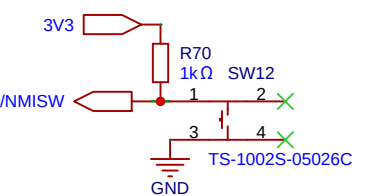
Soft Power Button



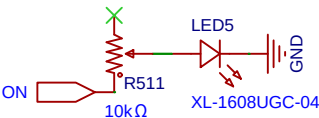
Reset Button



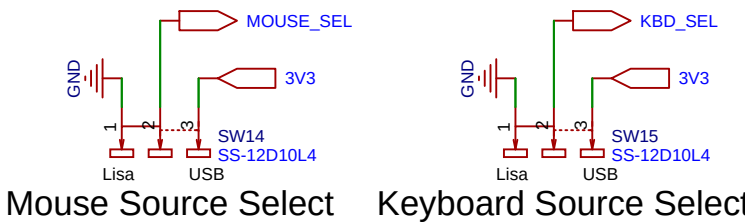
NMI Button



Soft Power LED

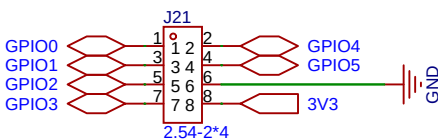


GPIO

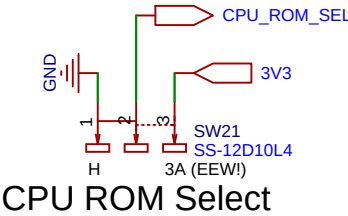


Mouse Source Select

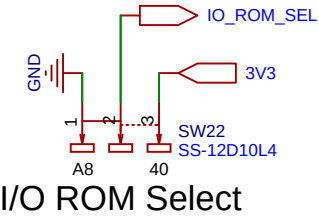
Keyboard Source Select



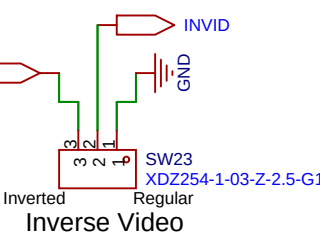
Config Switches



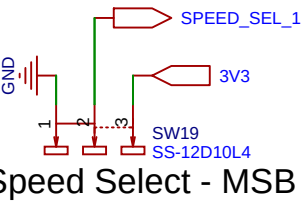
CPU ROM Select



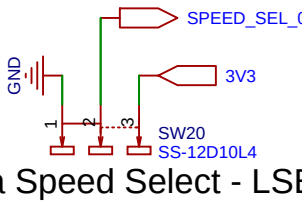
I/O ROM Select



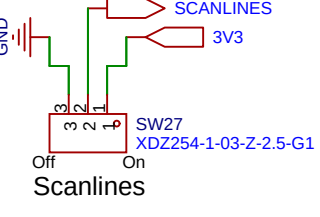
Inverse Video



Lisa Speed Select - MSB



Lisa Speed Select - LSB

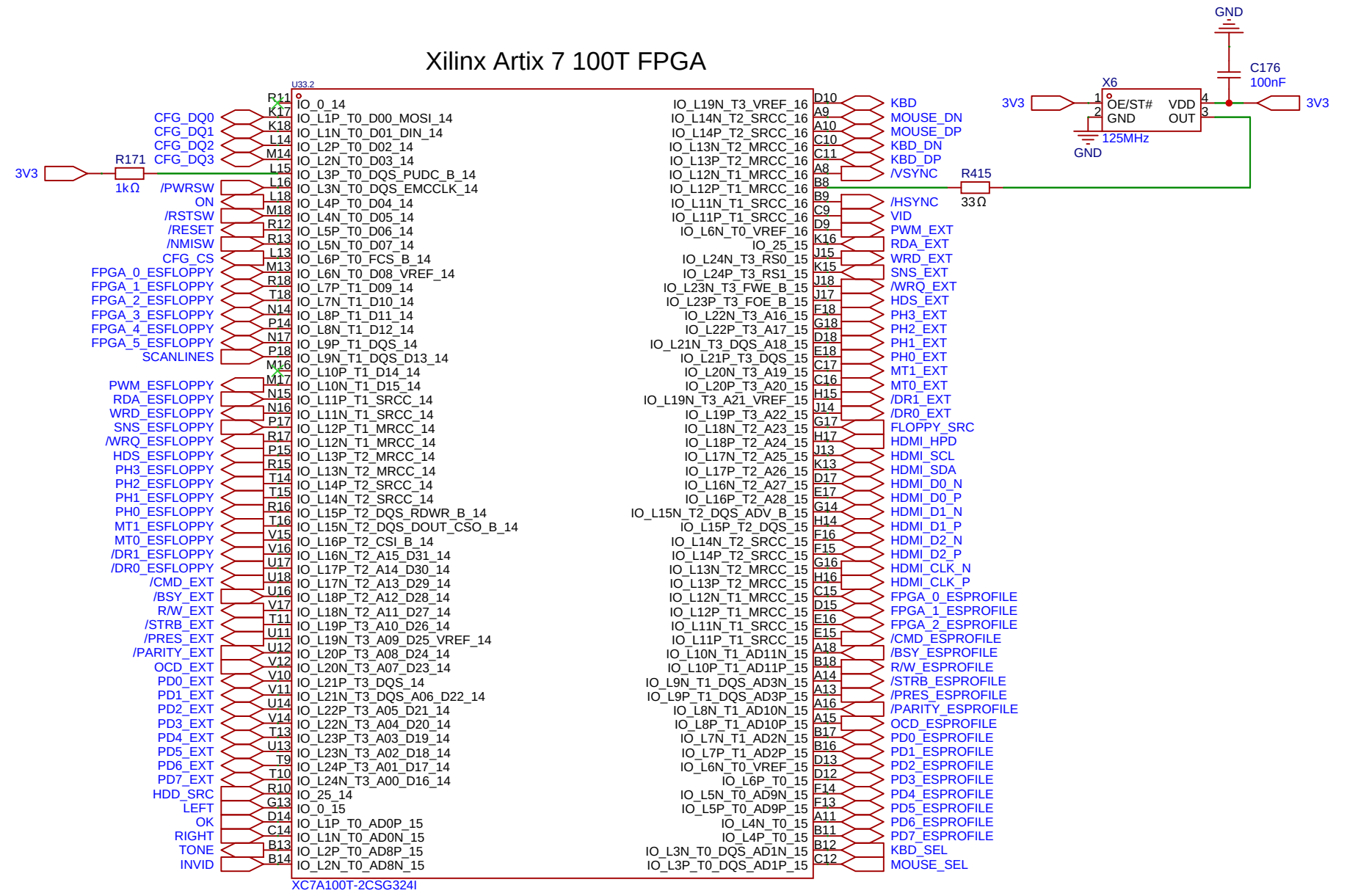


Scanlines

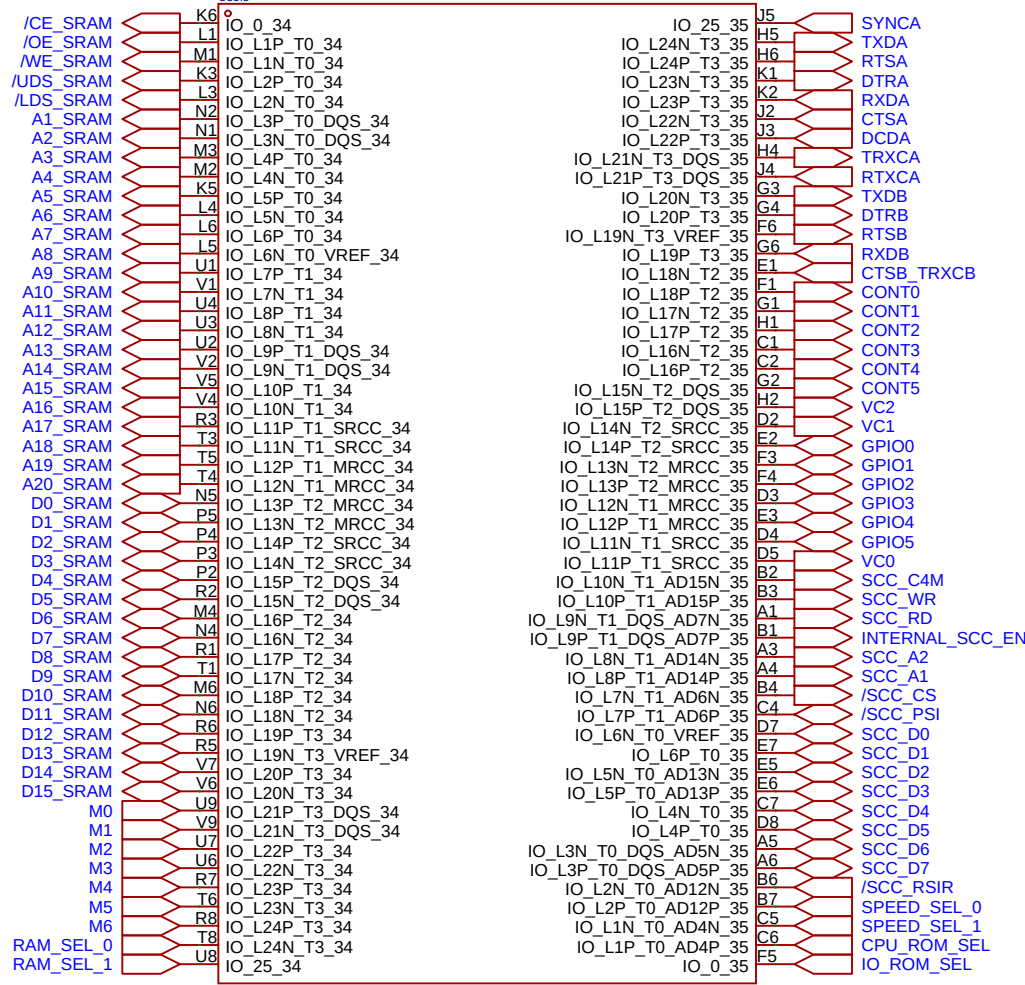
Wow, what a phenomenal use of space.
Looks like I screwed up my layout again...

Schematic	LisaFPGA Schematic			Create at	2025-10-28
Board	LisaFPGA			Update at	2026-03-12
Drawn				Page	Misc. I/O
Reviewed				LisaFPGA	
		Version	Size	Page 7 Total 8	
		V1.0	A4	EasyEDA.com	

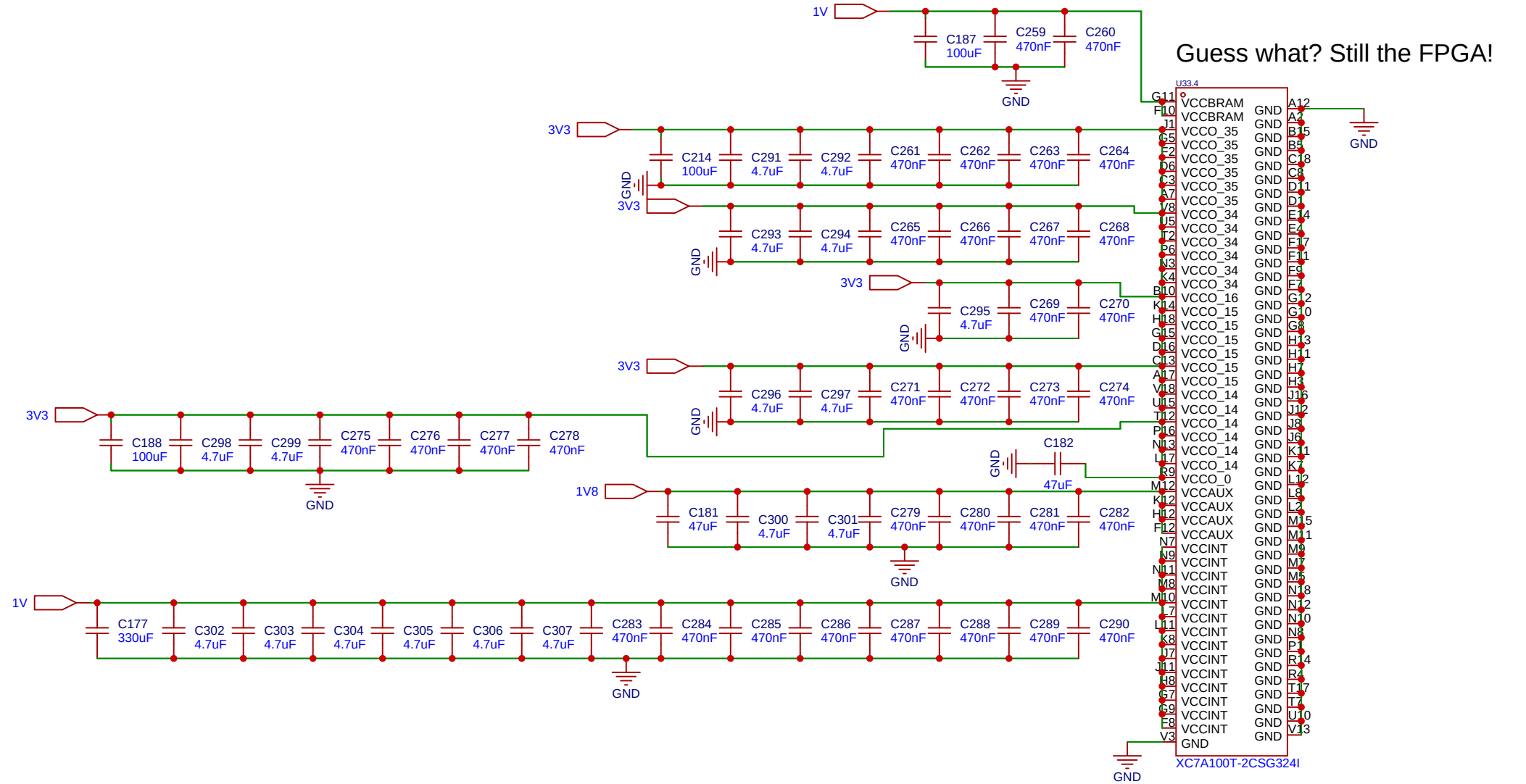
FPGA I/O Connections



The FPGA Again



FPGA Power



Schematic	LisaFPGA Schematic			Create at	2025-10-28
	LisaFPGA			Update at	2026-01-12
	LisaFPGA			Page	FPGA Connections
	LisaFPGA			Page 8 Total 8	
EasyEDA		Version	Size	EasyEDA.com	
		V1.0	A4		